

Core Pure Mathematics 1

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Chapter 2: Argand Diagrams

Why this matters

Representing complex numbers as points (or vectors) on a diagram turns algebra into geometry. This chapter builds the modulus-argument toolkit you need for later work with loci, and it gives you a visual, intuitive handle on what a complex number "is" — which pays off hugely in exam questions that mix algebra with sketching.

Concept: plotting complex numbers

An **Argand diagram** plots $z = a + bi$ as the point (a, b) , with the real part on the horizontal axis and the imaginary part on the vertical axis. Every complex number can equally be thought of as a position vector from the origin to that point.

Key vocabulary

Term	Meaning
Argand diagram	A coordinate plane with real axis (horizontal) and imaginary axis (vertical)
Modulus, z	The distance from the origin to z ; $ z = \sqrt{a^2 + b^2}$
Argument, $\arg(z)$	The angle (in radians, usually) measured anticlockwise from the positive real axis to the line Oz
Principal argument	The argument in the range $-\pi < \theta \leq \pi$
Modulus-argument (polar) form	$z = r(\cos\theta + i \sin\theta)$, where $r = z $ and $\theta = \arg(z)$
Locus	A path traced out by points satisfying a given condition

Unless a question specifically asks for a different range, always give the argument as the *principal* argument, $-\pi < \theta \leq \pi$.

Concept: modulus and argument

For $z = a + bi$, the modulus is $|z| = \sqrt{a^2 + b^2}$, found by Pythagoras' theorem on the Argand diagram. The argument is found using trigonometry on the right-angled triangle formed by z , its real part, and its imaginary part — but you must always sketch the point first to identify the correct quadrant, since a calculator's $\tan^{-1}$ only returns values between $-\pi/2$ and $\pi/2$.

Worked example 1: finding modulus and argument
EXAMPLE 1

Question. Find the modulus and argument of $z = -3 + 4i$, giving the argument in radians to 3 significant figures.

Step 1. Sketch the point $(-3, 4)$: it lies in the second quadrant (negative real, positive imaginary). **(B1)**

Step 2. Find the modulus: $|z| = \sqrt{(-3)^2 + 4^2} = \sqrt{25} = 5$. **(M1 A1)**

Step 3. Find the base angle using the positive values: $\alpha = \tan^{-1}(4/3) = 0.927$ radians (3 s.f.). **(M1)**

Step 4. Since z is in the second quadrant, $\arg(z) = \pi - \alpha = 3.14159... - 0.927 = 2.21$ radians (3 s.f.). **(A1)**

Answer: $|z| = 5$, $\arg(z) \approx 2.21$ radians.

Exam tip
EXAM TIP

The single biggest source of lost marks in this topic is skipping the sketch and trusting the calculator's raw $\tan^{-1}$ output. Make sketching the Argand diagram a non-negotiable first step, every single time — even for "obvious" cases — so the quadrant correction becomes automatic rather than an afterthought.

Concept: modulus-argument (polar) form

Once $r = |z|$ and $\theta = \arg(z)$ are known, z can be written as $z = r(\cos\theta + i \sin\theta)$. This form is especially useful for multiplying and dividing complex numbers geometrically (moduli multiply/divide, arguments add/subtract), and it underpins the loci work later in this chapter.

Worked example 2: converting to and from polar form
EXAMPLE 2

Question. Write $z = 2\sqrt{3} + 2i$ in the form $r(\cos\theta + i \sin\theta)$.

Step 1. Find the modulus: $r = \sqrt{((2\sqrt{3})^2 + 2^2)} = \sqrt{(12 + 4)} = \sqrt{16} = 4$. **(M1 A1)**

Step 2. The point is in the first quadrant, so $\theta = \tan^{-1}(2 / (2\sqrt{3})) = \tan^{-1}(1/\sqrt{3}) = \pi/6$. **(M1 A1)**

Answer: $z = 4(\cos(\pi/6) + i \sin(\pi/6))$.

Concept: loci — circles

The condition $|z - z_1| = r$ describes all points z whose distance from the fixed point z_1 is exactly r — that is, a circle of radius r centred at z_1 . This is because $|z - z_1|$ is literally the distance formula in disguise: if $z = x + iy$ and $z_1 = a + ib$, then $|z - z_1| = \sqrt{(x-a)^2 + (y-b)^2}$.

Worked example 3: circle locus
EXAMPLE 3

Question. Sketch the locus of points satisfying $|z - (3 - 2i)| = 4$, and describe it algebraically.

Step 1. Identify the fixed point: $z_1 = 3 - 2i$, plotted at $(3, -2)$. **(B1)**

Step 2. Recognise the condition as "distance from $(3, -2)$ equals 4", i.e. a circle centre $(3, -2)$, radius 4. **(M1 A1)**

Step 3. Sketch: draw the circle centre $(3, -2)$ with radius 4, clearly marking the centre and radius on the diagram. **(A1)**

Answer: The locus is the circle $(x - 3)^2 + (y + 2)^2 = 16$.

Concept: loci — half-lines

The condition $\arg(z - z_1) = \theta$ describes a **half-line** (ray) starting at z_1 (but not including z_1 itself, since $\arg$ is undefined there) and making an angle θ with the positive real direction. Because it is a half-line and not a full line, direction matters — you must draw an arrow and an open circle at the start point.

Worked example 4: half-line locus

EXAMPLE 4

Question. Sketch the locus given by $\arg(z - (1 + i)) = \pi/4$.

Step 1. Plot the start point $z_1 = 1 + i$ at $(1, 1)$ with an open circle, since it is excluded. **(B1)**

Step 2. From $(1, 1)$, draw a ray at angle $\pi/4$ (45°) to the positive real direction, extending outward with an arrow. **(M1 A1)**

Answer: A half-line starting (but not including) the point $(1, 1)$, at 45° to the real axis, extending up and to the right.

Concept: loci — perpendicular bisectors

The condition $|z - z_1| = |z - z_2|$ describes all points equidistant from z_1 and z_2 — by definition, the perpendicular bisector of the line segment joining z_1 and z_2 .

Worked example 5: perpendicular bisector locus

EXAMPLE 5

Question. Find the Cartesian equation of the locus $|z - 4| = |z - 2i|$.

Step 1. Let $z = x + iy$. Then $|z - 4| = \sqrt{(x-4)^2 + y^2}$ and $|z - 2i| = \sqrt{x^2 + (y-2)^2}$. **(M1)**

Step 2. Square both sides to remove the roots: $(x-4)^2 + y^2 = x^2 + (y-2)^2$. **(M1)**

Step 3. Expand: $x^2 - 8x + 16 + y^2 = x^2 + y^2 - 4y + 4$. **(A1)**

Step 4. Simplify: $-8x + 16 = -4y + 4$, which rearranges to $4y = 8x - 12$, i.e. $y = 2x - 3$. **(A1)**

Answer: The locus is the straight line $y = 2x - 3$ — the perpendicular bisector of the segment joining $(4, 0)$ and $(0, 2)$, as a quick check confirms.

Exam tip
EXAM TIP

Sanity-check every algebraic locus answer against a rough sketch of the two given points. Finding the midpoint of (4, 0) and (0, 2) — which is (2, 1) — and checking it satisfies $y = 2x - 3$ (it does: $2(2) - 3 = 1$) is a thirty-second check that catches most sign and expansion errors.

Concept: combining and interpreting loci

Exam questions often build on a basic locus in one of two ways:

- **Inequality regions**, e.g. $|z - z_1| \leq r$, meaning the interior and boundary of a disc, which must be shaded on the sketch.
- **Intersections of two loci**, e.g. a circle and a half-line, where you sketch both curves and use geometric reasoning — often right-angled triangles or the sine/cosine rule — to find greatest/least values of $|z|$ or $\arg(z)$.

Worked example 6: greatest and least value of $|z|$
EXAMPLE 6

Question. The complex number z satisfies $|z - (4 + 3i)| = 5$. Find the greatest and least possible values of $|z|$.

Step 1. Recognise the locus as a circle, centre (4, 3), radius 5. **(B1)**

Step 2. Find the distance from the origin O to the centre C : $|4 + 3i| = \sqrt{4^2 + 3^2} = \sqrt{25} = 5$. **(M1 A1)**

Step 3. Sketch the circle together with the line through O and C . Since $OC = 5$ equals the radius, the origin itself lies on the circle, so the least value of $|z|$ is 0. The greatest value occurs at the point on the circle furthest from O , along the extension of OC beyond C : greatest $|z| = OC + \text{radius} = 5 + 5 = 10$. **(M1 A1)**

Answer: The least value of $|z|$ is 0 and the greatest value is 10.

Exam tip
EXAM TIP

The general pattern for greatest/least $|z|$ on a circle $|z - z_1| = r$ is: greatest $|z| = |z_1| + r$, least $|z| = ||z_1| - r|$, found by sketching the straight line from the origin through the centre. Always sketch this line rather than reasoning about it purely algebraically — the picture makes the addition or subtraction obvious, and shows immediately whether the origin lies inside, outside, or on the circle.

Exactly the same picture handles greatest/least **argument** questions: for the circle $|z - z_1| = r$, sketch the two tangent lines from the origin to the circle. The greatest and least values of $\arg(z)$ occur where these tangents touch the circle, and can be found using $\sin(\text{angle}) = r / |z_1|$ together with $\arg(z_1)$, since the tangent, the radius to the point of contact, and the line from O to z_1 form a right-angled triangle.

Chapter summary

Idea	Key point
Argand diagram	Plots $z = a + bi$ as the point (a, b)
Modulus	$ z = \sqrt{a^2 + b^2}$, the distance from the origin
Argument	Angle from positive real axis; always sketch first to fix the quadrant
Polar form	$z = r(\cos\theta + i \sin\theta)$
Circle locus	$ z - z_1 = r$ is a circle, centre z_1 , radius r
Half-line locus	$\arg(z - z_1) = \theta$ is a ray from (excluding) z_1 at angle θ
Perpendicular bisector locus	$ z - z_1 = z - z_2 $ is the perpendicular bisector of z_1z_2

Common exam mistakes

- Using the calculator's raw $\tan^{-1}$ value without adjusting for the correct quadrant.
- Giving the argument in the wrong range (should usually be $-\pi < \theta \leq \pi$ unless told otherwise).
- Drawing a full line instead of a half-line (forgetting the arrow and open circle) for $\arg(z - z_1) = \theta$.
- Forgetting that the point z_1 itself is excluded from a half-line locus.
- Sign errors when expanding brackets while finding a perpendicular bisector equation.
- Not shading or clearly indicating the correct region for inequality loci.

You should now be able to

- Plot complex numbers on an Argand diagram.
- Calculate the modulus and argument of a complex number, correcting for quadrant.
- Convert between Cartesian and modulus-argument (polar) form.
- Sketch and derive the Cartesian equation of circle, half-line, and perpendicular bisector loci.
- Interpret regions defined by locus inequalities and find greatest/least values of $|z|$ or $\arg(z)$.